

CS-411: Artificial Intelligence

Time: 1.30 Hrs

Max. Marks: 50

Note:

- Attempt all questions.
- Assume missing data suitably, if any.
- No answer should be fragmented at difference places.

- Q1. Write brief notes on the following in respect to Historic perspective of AI: 12
- a. Program, possibly the first one that could learn playing a game.
 - b. Expert system, possibly the first one that could replicate the ability of human experts.
 - c. Autonomous drawing program possibly the first one.
 - d. Program, possibly the first one that could simulate the human emotions.
- Q2. What do we mean by an agent and its rationality? Discuss the framework to specify the structure of an intelligent agent. 10
- Q3. What do we mean by a problem and the problem solving especially in AI? Discuss its different aspects. Write pseudocode for a problem solving agent. 10
- Q4. The searching techniques are generally discussed in Data Structures/algorithms. Why are they relevant in AI? Write the pseudocode of the best first search approach. How can we develop different search techniques that mainly use the best first search approach? 10
- Q5. Discuss the basic concept of the hill climbing search in AI and related concepts/terms. 8

CS-411: Artificial Intelligence

Time: 3.00 Hrs

Max. Marks: 50

Note:

- Attempt all questions, each carries equal marks. In case two parts, each has equal marks.
- Assume missing data suitably, if any.
- No answer should be fragmented at difference places.

Q1(a). What do you mean by heuristic and what are possible ways to find the heuristic value? Apply any heuristic based method to solve the adjacent problem.

Does any initial state always lead to a goal state? Discuss.

Initial state

Goal state

	1	3
4	2	5
7	8	6

1	2	3
4	5	6
7	8	

(b) Write the simulating annealing algorithm and apply it to the adjacent problem.

Q2. Write short notes on the following by giving suitable examples:

- a) Completeness; b) Soundness; c) Resolution; d) Unification; e) Skolemization.

Q3(a). Write the alpha-beta pruning algorithm and apply it for the tree like problem given below.

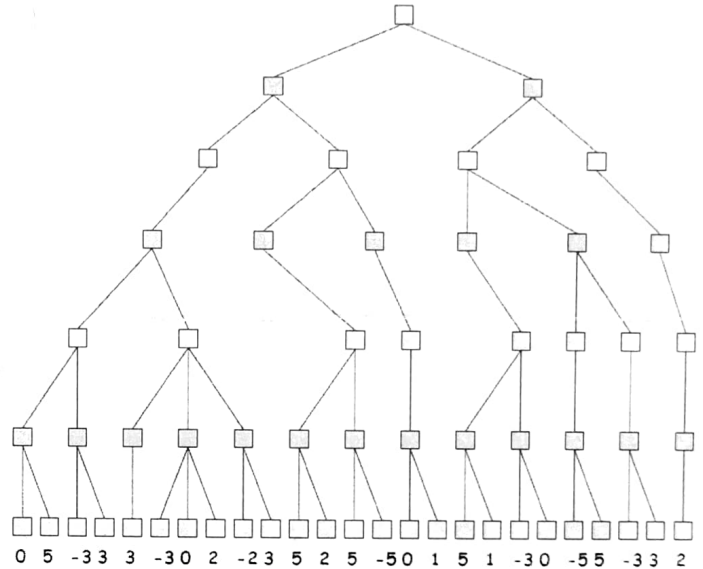
(b). Write the axioms for defining the natural numbers.

Q4(a). Why do we need conceptual dependency?

Write its primitives.

(b) Write following sentences using these primitives:

- i. Yesterday, Rohit hit his BMW white care.
- ii. Mohit is giving water to a mango plant.
- iii. Hari is sad because Sohan beat him
- iv. When Lina gave Mira a peach, she ate it.
- v. Deepak will fly his plane to Mumbai tomorrow.

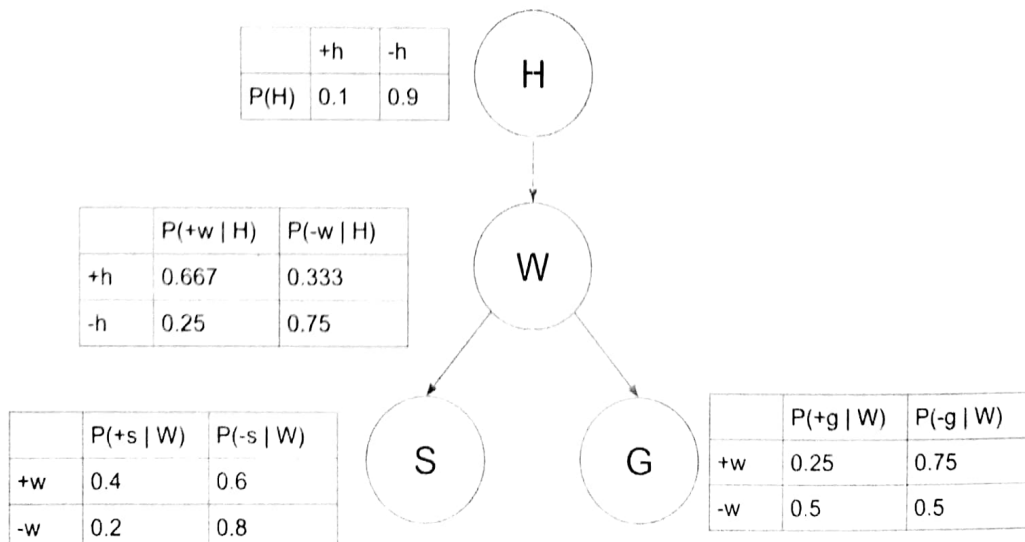


Q5(a) The Modus ponens is generally used to decide the truth value of a statement based on two statements. Discuss this rule for the probabilistic domain.

When does this rule coincide in logic and probabilistic domains?

(b) What do you mean by the Bayesian networks? Calculate the joint distribution of the random variables used in the following network:

	+h	-h
P(H)	0.1	0.9



* * * * *